Day 3 of 15 | Core Unit: Apply Statistical Concepts and Analysis

Probability, Correlation, Regression and the t-Test

Student Handout · TESDA Data Analytics Level III · Batch DAL3-2026

Case company: **Tindahan ni Aling Nena Mini-Mart** · payments, rice-bag weights, hot-weather drink sales, branch comparisons

Learning Outcomes

By the end of Day 3 you should be able to:

1. Compute and interpret a **probability**, and use the complement and "or" rules.
2. Describe the **normal distribution**, compute **z-scores**, and use `NORM.DIST` to find a probability.
3. Measure the relationship between two variables with **correlation (r)**, and show it in a **scatter chart**.
4. Fit a **simple linear regression** line and use it to **forecast**.
5. Use a **t-test** to decide whether the difference between two averages is **real or just chance**, by reading the **p-value**.
6. Cross-check with the **Analysis ToolPak**, and explain results in plain words.

Materials you need today

Item	File / Location
Activity workbook	`Day03/assets/Day03_Analysis_Template.xlsx`
Demo video (with captions, shortcut-key overlays)	`Day03/demo/Day03_Demo_Probability_Correlation_Regression_t-Test.mp4`
Activity sheets	`Day03_Student_Activities.pdf`

> **How to use this handout.** Every new term has a **"Sa simpleng salita"** (in simple words) box. Read that first, then the formal definition. If a word is still unclear, check the **Glossary** on the last page.

Part 1 — Probability: How Likely Is It?

> **Sa simpleng salita:** Probability is *how likely* something is to happen. A jar has 10 candies and 6 are orange. If you pick one without looking, the chance of orange is 6 out of 10 = **0.6 = 60%**.

Probability of an event A

P(A) = number of favorable outcomes ÷ total number of outcomes

Value	Meaning
0 (0%)	Impossible: it will never happen
0.5 (50%)	As likely as not, like a coin toss
1 (100%)	Certain: it will always happen

Worked example (Day 1 data, 500 transactions): 151 customers paid with GCash.
P(GCash) = 151 ÷ 500 = **0.302 = 30.2%**. In plain words: *about 3 of every 10 customers pay with GCash.*

Three simple rules

Rule	Formula	Example (500 transactions)
All probabilities add up to 1	P(Cash) + P(GCash) + P(Maya) + P(Card) = 1	0.560 + 0.302 + 0.064 + 0.074 = 1.000
Complement ("not A")	P(not A) = 1 − P(A)	P(not Cash) = 1 − 0.560 = **0.440**
"Or" (events that cannot happen together)	P(A or B) = P(A) + P(B)	P(GCash or Maya) = 0.302 + 0.064 = **0.366**

Expected count = n × P(A). Out of the next 200 customers, expect about 200 × 0.302 ≈ **60 GCash payers**, so make sure the GCash QR stand is visible.

In Excel: `=COUNTIF($C$2:$C$501,"GCash")/COUNTA(C2:C501)`

Part 2 — Quick Review: Standard Deviation

> **Sa simpleng salita:** The standard deviation (SD) is the *typical distance* of the values from the average. Two stores both average ₱100 an hour. Store A always sells ₱95–₱105 (**small SD**, consistent). Store B sells ₱60 one hour and ₱140 the next (**big SD**, unpredictable).

`=STDEV.S(range)` gives the sample standard deviation (Day 2). You need it for z-scores today.

Part 3 — The Normal Distribution and z-Scores

The normal distribution (bell curve)

> **Sa simpleng salita:** Think of the heights of all students in a school. Most are of average height, a few are very tall, and a few are very short. If you graph it, it looks like a **bell**: high in the middle, low at both ends.

Many real measurements are approximately **normal**: the weight of packed rice bags, travel time, test scores. A normal distribution is described by two numbers only: its **mean** (center) and its **standard deviation**

(spread).

The 68–95–99.7 rule (empirical rule)

Range	Share of the data
mean ± 1 SD	about **68%**
mean ± 2 SD	about **95%**
mean ± 3 SD	about **99.7%**

z-score: how many SDs away?

> **Sa simpleng salita:** A z-score tells you *how many standard deviations* a value is from the average. z = 0 means exactly average. z = −2.5 means "2.5 SDs **below** average". That is very unusual.

z = (value − mean) ÷ standard deviation Excel: `=(B2-$G$3)/$G$4` or `=STANDARDIZE(x, mean, sd)`

Worked example: 5-kg rice bags. Mean = 5.04 kg, SD = 0.04 kg.

Bag	Weight	z	Interpretation
A	5.04 kg	(5.04 − 5.04) ÷ 0.04 = **0**	Exactly average
B	5.12 kg	(5.12 − 5.04) ÷ 0.04 = **+2.0**	2 SDs above, on the edge of unusual
C	4.95 kg	(4.95 − 5.04) ÷ 0.04 = **−2.25**	Unusual, and **underweight**: check the scale

Rule of thumb: a value with **z > +2 or z < −2** is **unusual**. Investigate it, the same way we treated outliers on Day 2.

Probability from the normal curve: `NORM.DIST`

`=NORM.DIST(x, mean, sd, TRUE)` gives the probability of a value **less than or equal to x**.
Example: P(bag < 5.000 kg) = `=NORM.DIST(5, 5.04, 0.0397, TRUE)` ≈ **0.157 = 15.7%**.
In plain words: *about 157 of every 1,000 bags would be under 5 kg.* Selling underweight goods breaks the Consumer Act, so the packing scale must be adjusted.

You want	Excel
P(X ≤ x)	`=NORM.DIST(x, mean, sd, TRUE)`
P(X > x)	`=1-NORM.DIST(x, mean, sd, TRUE)`
P(a ≤ X ≤ b)	`=NORM.DIST(b,…,TRUE)-NORM.DIST(a,…,TRUE)`

Part 4 — Correlation: Do Two Things Move Together?

> **Sa simpleng salita:** Correlation tells us if two things **move together**. When the weather is hotter, more people buy cold drinks: both go **up together** (positive). When the price of pancit canton goes up, fewer packs are sold: one goes up and the other goes **down** (negative).

Pearson correlation coefficient r ranges from **−1 to +1**.

r	Strength and direction	Scatter chart looks like
+0.7 to +1.0	Strong positive	Dots rising tightly from left to right
+0.3 to +0.7	Moderate positive	Rising, but scattered
−0.3 to +0.3	Weak or none	A cloud with no pattern
−0.7 to −0.3	Moderate negative	Falling, but scattered
−1.0 to −0.7	Strong negative	Dots falling tightly from left to right

Excel: `=CORREL(x_range, y_range)`. Make a **scatter chart**: select both columns → **Insert** → **Scatter**.

Worked example (5 hot days):

Max temp °C (x)	Cold drinks sold (y)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
30	110	−3	−36.6	109.8
31	118	−2	−28.6	57.2
33	150	0	3.4	0
35	170	2	23.4	46.8
36	185	3	38.4	115.2
$\bar{x} = 33$	$\bar{y} = 146.6$	$\Sigma(x - \bar{x})^2 = 26$	$\Sigma(y - \bar{y})^2 = 4{,}191.2$	**Σ = 329**

$r = \Sigma(x - \bar{x})(y - \bar{y}) \div \sqrt{[\Sigma(x - \bar{x})^2 \times \Sigma(y - \bar{y})^2]} = 329 \div \sqrt{(26 \times 4{,}191.2)} = 329 \div 330.11 =$ **0.997** → very strong positive.

> ⚠️ **Correlation is not causation.** Ice-cream sales and drowning cases both rise in summer. Ice cream does not cause drowning; **hot weather** drives both. Always ask: is there a third factor? *(Integrated: Apply critical thinking.)*

Part 5 — Simple Linear Regression: Draw a Line, Then Forecast

> **Sa simpleng salita:** Regression draws the **one straight line that is closest to all the dots**. Then we use the line to **guess (forecast)**. If PAGASA says it will be 37 °C tomorrow, the line tells us how many cold drinks to prepare.

$\hat{y} = a + b \cdot x$

Part	Meaning	Excel
b, the slope	How much y changes when x goes up by 1	`=SLOPE(known_y, known_x)`
a, the intercept	Where the line starts (the value of ŷ when x = 0)	`=INTERCEPT(known_y, known_x)`
R^2	Share of the change in y explained by x (0 to 1)	`=RSQ(known_y, known_x)`
Forecast	The predicted y for a new x	`=FORECAST.LINEAR(x, known_y, known_x)`

⚠️ In Excel the **y range comes first**, then the x range: `SLOPE(y, x)`. That's the opposite of how we usually say "x and y".

Worked example (same 5 days): b = 329 ÷ 26 = **12.65** drinks per °C; a = 146.6 − 12.65 × 33 = **−270.98**. Forecast for 34 °C: ŷ = −270.98 + 12.65 × 34 = **159 drinks**.

Today's full dataset (60 days): slope ≈ **12.28**, intercept ≈ −258.01, R^2 ≈ **0.76**, forecast at 37 °C ≈ **196 drinks**.
In plain words: *each extra degree of heat sells about 12 more cold drinks, and temperature explains about three-quarters of the ups and downs in drink sales.*

Add the line to a chart: select the chart → **Chart Design** → **Add Chart Element** → **Trendline** → **Linear**. (More Trendline Options lets you tick *Display Equation* and *Display R-squared*.)

> **Don't forecast too far outside your data.** Our data covers 28.6–38.9 °C. A forecast for 45 °C is a guess outside what we have seen.

Part 6 — Hypothesis Testing, the t-Test and the p-Value

> **Sa simpleng salita:** Is the difference **real**, or just **tsamba** (luck)? Toss a fair coin 10 times and get 6 heads: that is normal luck. But 10 heads out of 10? That is so unlikely for a fair coin that you'd suspect the coin is not fair. The **p-value** measures exactly that: *how likely a result this big would be if it were only luck.*

The steps of a hypothesis test

1. **Null hypothesis (H_0):** "There is no real difference." Example: Marikina and Pasig have the same average daily sales.
2. **Alternative hypothesis (H_1):** "There is a real difference."
3. Choose the **significance level α = 0.05** (5%).
4. Run the test and get the **p-value**.
5. **Decide:** if **p < 0.05**, reject H_0: the difference is **statistically significant** (real). If **p ≥ 0.05**, we **cannot** say it is real; it could be chance.

The two-sample t-test in Excel

`=T.TEST(range1, range2, tails, type)` returns the **p-value**.

Argument	Use	Why
tails = **2**	Two-tailed	We ask "different?", not "higher?"
type = **3**	Two-sample, **unequal variances** (Welch)	The safe default when spreads differ
type = 2	Two-sample, equal variances	Only if the SDs are about the same
type = 1	Paired	The same items measured twice (before/after on the same store-days)

Today's two results

Test	Averages	p-value	Decision
Marikina vs Pasig daily sales (90 days each)	₱14,994 vs ₱16,644	**0.00004**	p < 0.05 → **real difference**: Pasig really sells more
Caloocan basket before vs after the new shelf layout (25 days each)	₱92.36 vs ₱95.47	**0.274**	p ≥ 0.05 → **not proven**: it could be chance; collect more data

> **Read p-values correctly.** A big p-value does **not** prove "no effect". It only says the evidence is **not strong enough yet**. And "significant" means "probably not luck", not "big" or "important". Always look at the size of the difference too.

Cross-check with the Analysis ToolPak

Data → Data Analysis → t-Test: Two-Sample Assuming Unequal Variances → OK → Variable 1 Range `$B$1:$B$91`, Variable 2 Range `$C$1:$C$91`, ✓ **Labels**, **New Worksheet Ply** → OK.
Read **P(T<=t) two-tail**: it must equal your `T.TEST` result. *(Integrated: Apply quality standards.)*

Shortcut Keys Used Today

Keys	What it does
Ctrl + ↓ / **Ctrl** + ↑	Jump to the last / first filled cell in a column
Ctrl + **Home**	Go to the top-left of the sheet
Ctrl + **Shift** + ↓	Select from the current cell down to the last filled cell
Shift + ↓ / →	Extend the selection one cell
F4 (while typing a formula)	Add `$` signs to the reference you just typed (lock it). Press again to cycle `$A$1 → A$1 → $A1 → A1`
Ctrl + **D** / **Ctrl** + **R**	Fill down / fill right from the first cell of the selection

Keys	What it does
Double-click the **fill handle**	Copy a formula down as far as the next column has data
Alt + =	AutoSum
Ctrl + Shift + %	Percent style
Ctrl + Page Down / Page Up	Next / previous sheet
Ctrl + S	Save

Excel Function Summary

Purpose	Function
Count matches / count non-empty	`=COUNTIF(r, criteria)` · `=COUNTA(r)`
z-score	`=(x-mean)/sd` · `=STANDARDIZE(x, mean, sd)`
Normal probability P(X ≤ x)	`=NORM.DIST(x, mean, sd, TRUE)`
Correlation	`=CORREL(x_range, y_range)`
Regression	`=SLOPE(y, x)` · `=INTERCEPT(y, x)` · `=RSQ(y, x)`
Forecast	`=FORECAST.LINEAR(new_x, known_y, known_x)`
t-test p-value	`=T.TEST(r1, r2, 2, 3)`

Glossary

Term	Formal definition	Sa simpleng salita
Probability	Favorable outcomes ÷ total outcomes, from 0 to 1	Gaano ka-posible na mangyari
Complement	P(not A) = 1 − P(A)	Ang chance na *hindi* mangyari
Normal distribution	A symmetric, bell-shaped distribution set by its mean and SD	Hugis-kampana: karamihan nasa gitna
Empirical rule	≈68% within ±1 SD, ≈95% within ±2 SD, ≈99.7% within ±3 SD	Halos lahat ay nasa loob ng ±2 SD
z-score	(value − mean) ÷ SD	Ilang SD ang layo sa average
Correlation (r)	Strength and direction of a straight-line relationship, −1 to +1	Sabay bang gumagalaw ang dalawa?
Causation	One variable directly causes the change in another	Ito mismo ang dahilan (iba sa correlation)

Term	Formal definition	Sa simpleng salita
Regression line	$\hat{y} = a + b \cdot x$, the line of best fit	Ang linyang pinakamalapit sa mga tuldok
Slope (b)	Change in y for each +1 in x	Dagdag sa y kada +1 sa x
Intercept (a)	Value of $\hat{y}$ when x = 0	Simula ng linya
R^2	Proportion of the variation in y explained by x	Gaano kalaki ang naipapaliwanag ng x
Forecast	A predicted value from the model	Hula batay sa linya
Null hypothesis (H_0)	The claim of "no difference / no effect"	"Walang totoong difference"
p-value	Probability of a result at least this extreme if H_0 were true	Chance na tsamba lang ito
Significance level (α)	The cut-off for deciding, usually 0.05	Ang hangganan: 5%
Statistically significant	$p < \alpha$; the difference is unlikely to be chance	Malamang totoo, hindi tsamba
t-test	A test comparing the means of two groups	Ikumpara ang average ng dalawang grupo

Key Takeaways

- Probability is between 0 and 1. All outcomes add up to 1, and "not A" = 1 − P(A).
- A z-score beyond ±2 means a value is unusual. Investigate it.
- r measures how strongly two variables move together. **Correlation ≠ causation.**
- A regression line turns a relationship into a **forecast**. Check R^2, and don't forecast far outside your data.
- **$p < 0.05$ → the difference is real; $p \geq 0.05$ → not proven yet.** Cross-check with the ToolPak.

Next session (Day 4): Prepare Data Sets — introduction to databases and SQL with VS Code and SQLTools.